

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent Application Publication

20250278101

Kind Code

A1

Publication Date

September 04, 2025

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MANUAL CONTROL RE-ENGAGEMENT IN AN AUTONOMOUS VEHICLE

Abstract

Vehicles may have the capability to navigate according to various levels of autonomous capabilities, the vehicle having a different set of autonomous competencies at each level. In certain situations, the vehicle may shift from one level of autonomous capability to another. The shift may require more or less driving responsibility from a human operator. Sensors inside the vehicle collect human operator parameters to determine an alertness level of the human operator. An alertness level is determined based on the human operator parameters and other data including historical data or human operator-specific data. Notifications are presented to the user based on the determined alertness level that are more or less intrusive based on the alertness level of the human operator and on the urgency of an impending change to autonomous capabilities. Notifications may be tailored to specific human operators based on human operator preference and historical performance.

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Family ID: 74679489

Appl. No.: 19/209467

Filed: May 15, 2025

Related U.S. Application Data

parent US continuation 16950588 20201117 parent-grant-document US 12326727 child US 19209467

parent US division 15957637 20180419 parent-grant-document US 10935974 child US 16950588

Publication Classification

Int. Cl.: G05D1/00 (20240101); B60W50/00 (20060101); G05D1/81 (20240101)

U.S. Cl.:

CPC G05D1/81 (20240101); B60W2050/007 (20130101); B60W2050/0072 (20130101)

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION [0001] This application is a continuation of U.S. application Ser. No. 16/950,588, filed on Nov. 17, 2020, entitled “Manual Control Re-Engagement in an Autonomous Vehicle,” which is a divisional of U.S. application Ser. No. 15/957,637, filed on Apr. 19, 2018, entitled “Manual Control Re-Engagement in an Autonomous Vehicle.” The entirety of each of the foregoing applications is expressly incorporated herein by reference.

BACKGROUND

[0002] Vehicles may be equipped with sensors and processing and communication capabilities that allow the vehicle to navigate autonomously without human intervention. Autonomous vehicle navigation is not possible, however, under all circumstances. In some situations, a vehicle may lack the capability to navigate autonomously such as adverse or extreme weather conditions, in the event of the loss of vehicle sensors or a communications link, under disaster conditions, due to vehicle collisions in the area, etc. When a vehicle can no longer navigate autonomously, control of the vehicle may be returned to a human operator. Transfer of control of the vehicle to a human represents a potential danger due to the vehicle's autonomous nature—a human operator who has been traveling in the vehicle may be inattentive to road conditions and unprepared to make the decisions and exercise the motor control needed to safely operate the vehicle.

SUMMARY OF THE DISCLOSURE

[0003] This summary is provided to introduce a selection of concepts in a simplified form that are further described in the Detailed Descriptions. This summary is not intended to identify key features or essential features of the claimed subject matter, nor is it intended to be used to limit the scope of the claimed subject matter.

[0004] Systems and methods are disclosed for requesting engagement of manual control in a vehicle. The vehicles may have the capability to navigate according to various levels of autonomous capabilities, where the vehicle has a different set of autonomous competencies at each level. In certain situations, the vehicle may shift from one level of autonomous capability to another. The shift may require more or less driving responsibility from a human operator. Sensors inside the vehicle may collect human operator parameters to determine an alertness level of the human operator. An alertness level may be determined based on the human operator parameters and other data including historical data or human operator-specific data. Notifications may be presented to the user based on the determined alertness level that are more or less intrusive based on the alertness level of the human operator and on the urgency of an impending change to autonomous capabilities. Notifications may be tailored to specific human operators based on human operator preference and historical performance.

[0005] In one aspect, a method of requesting engagement of manual control in a vehicle is disclosed. The method may include identifying a human operator of the vehicle, the human operator being associated with a driving capability profile; determining human operator parameters associated with the human operator of the vehicle; and determining an alertness level of the human

operator based on the human operator parameters and the driving capability profile. The method may also include receiving an indication to transfer control of the vehicle to the human operator; displaying a manual control request notification to the human operator, the manual control request notification communicating a request for manual control, the manual control request notification further being based on the determined alertness level; and transferring control of the vehicle to the human operator if the human operator accepts the request for manual control.

[0006] In another aspect, a system for transferring control of a vehicle to a human operator is disclosed. The system may include an evaluator configured to evaluate readiness of a human operator of a vehicle, the evaluator including one or more sensors; a vehicle security arbiter configured to determine a security threat to the vehicle; and a notification server configured to display a first manual control request notification to the human operator, the first manual control request notification being based at least in part on the readiness of the human operator and the security threat to the vehicle. The notification server being further configured to display a second manual control request notification to the human operator, the second manual control request notification being different from the first manual control request notification; and the vehicle security arbiter being further configured to transfer control of the vehicle to the human operator.

[0007] In yet another aspect, a method is disclosed for transitioning control of a vehicle. The method may include navigating a vehicle according to a first level of autonomous capability, the first level of autonomous capability defining a first set of autonomous competencies of the vehicle; receiving a request to operate the vehicle according to a second level of autonomous capability within a time period, the second level of autonomous capability defining a second set of autonomous competencies, the second set of autonomous competencies being different than the first set of autonomous competencies; and notifying a human operator of the vehicle with a notification of an impending change in vehicle capability to the second level of autonomous capability, the notification depending at least in part on differences between the first set of autonomous competencies and the second set of autonomous competencies. The method may also include navigating the vehicle according to the second level of autonomous capability.

[0008] The methods may be implemented via computer systems, and may include additional, less, or alternate actions or functionality. Systems or computer-readable media storing instructions for implementing all or part of the method described above may also be provided in some aspects. Systems for implementing such methods may include one or more of the following: a special-purpose computing device, a personal electronic device, a mobile device, a wearable device, a processing unit of a vehicle, a remote server, one or more sensors, one or more communication modules configured to communicate wirelessly via radio links, radio frequency links, and/or wireless communication channels, and/or one or more program memories coupled to one or more processors of the personal electronic device, processing unit of the vehicle, or remote server. Such program memories may store instructions to cause the one or more processors to implement part or all of the method described above. Additional or alternative features described herein below may be included in some aspects.

[0009] This summary is provided to introduce a selection of concepts in a simplified form that are further described in the Detailed Descriptions. This summary is not intended to identify key features or essential features of the claimed subject matter, nor is it intended to be used to limit the scope of the claimed subject matter.

[0010] Advantages will become more apparent to those of ordinary skill in the art from the following description of the preferred aspects, which have been shown and described by way of illustration. As will be realized, the present aspects may be capable of other and different aspects, and their details are capable of modification in various respects. Accordingly, the drawings and description are to be regarded as illustrative in nature and not as restrictive.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0011] FIG. 1 is a schematic diagram of an example system for manual control re-engagement in a vehicle.

[0012] FIG. 2 is a time-series schematic diagram of an example system for manual control re-engagement in a vehicle with a human occupant.

[0013] FIG. 3 is an in-vehicle view of an example system for manual control re-engagement in a vehicle.

[0014] FIG. 4 is an in-vehicle view of an example system for manual control re-engagement in a vehicle.

[0015] FIG. 5 is an in-vehicle view of an example system for manual control re-engagement in a vehicle.

[0016] FIG. 6 is an in-vehicle view of an example system for manual control re-engagement in a vehicle.

[0017] FIG. 7 is an in-vehicle view of an example system for manual control re-engagement in a vehicle.

[0018] FIG. 8 is a schematic diagram of a vehicle equipped with an example system for manual control re-engagement in a vehicle approaching a road hazard.

[0019] FIG. 9 is an in-vehicle view of an example system for manual control re-engagement in a vehicle.

[0020] FIG. 10 illustrates example operations for requesting engagement of manual control in a vehicle.

[0021] FIG. 11 illustrates example operations for transitioning control of a vehicle.

DETAILED DESCRIPTIONS

[0022] Autonomous vehicles may exercise a range of capabilities when navigating on open road conditions. An autonomous vehicle need not be viewed as operating purely autonomously or purely manually. The Society of Automotive Engineers (SAE) has identified at least six levels of autonomous vehicle capability ranging from no driving automation (Level 0) to full automation (Level 5). As a vehicle moves up the levels of capability, additional autonomous competencies are added to the vehicle's set of skills (e.g., adaptive cruise control, parking assist, lane assist, traffic jam assist, conditional automation, high automation, full automation, etc.).

[0023] At the various points on the vehicle's autonomous skill ladder, a human operator exercises an appropriate level of manual control. If a vehicle supplies only adaptive cruise control or parking assist capabilities, then the human operator must exercise a high level of manual control and is responsible for any and all non-autonomous aspects of the vehicle. If a vehicle supplies high or full automation, on the other hand, a human operator may participate at a low level, or even not at all, in navigating the vehicle. If a vehicle exercises a high level of autonomous capability (e.g., Levels 4 or 5), a human operator may become disengaged from the road and unaware of the road environment surrounding the vehicle. The human user may become focused on non-driving tasks (e.g., reading, working, playing games, conversations with other passengers, phone calls, etc.). A human operator may even go to sleep and lose all contact with controlling the vehicle.

[0024] It may be desirable for an autonomous vehicle to shift from one level of autonomous capability to another, such as due to changing road conditions, weather conditions, due to a vehicle crash, disaster or emergency situation, etc. To the extent the human operator will need to change her involvement in piloting the vehicle when the vehicle shifts between levels of automation, she must be notified of the impending change. A variety of notification types are used in the system for manual control re-engagement to communicate with the human operator regarding upcoming changes in the human operator's responsibilities in piloting the vehicle. A notification may be more

intrusive to the human operator or less intrusive to the human operator depending on the urgency of the impending change to the vehicle's capabilities and the human operator's responsibilities.

[0025] Different human operators will respond differently to notifications of changes to the vehicle's autonomous capabilities. Human operators exhibit differences from one another in terms of attention span, ability to multitask, ability to shift focus from one activity to another after receiving a notification, etc. To some extent, these differences are natural characteristics of the human operators themselves (e.g., some people are more likely than others to become engrossed in reading a book and may take more time to respond to a notification of impending vehicle autonomous capability change than other people who may tend not to become engrossed in reading a book while riding in a car). In other situations, a human operator's ability to acknowledge a notification and prepare to exercise more or less control over the vehicle may depend on other aspects of the human operator that may change over time (e.g., intoxication level, how well rested the human operator is, whether the human operator has become focused on another activity or remains aware of the vehicle's surroundings, the health of the human operator, etc.). In yet other situations, a human operator's ability to acknowledge a notification and prepare to exercise more or less control over the vehicle may change slowly over time (e.g., a human operator's eyesight may deteriorate over time, motor control deteriorates with age, etc.).

[0026] A change to a vehicle's autonomous capability may also have a variable time target in which the change should be made. Some changes to a vehicle's autonomous capabilities must be made quickly, such as in emergency situations (e.g., a vehicle crash or other rapidly approaching road hazard, if the human operator experiences an emergent medical or health problem, etc.). Other changes to a vehicle's autonomous capabilities need not be made quickly (e.g., if adverse or extreme weather conditions are forecasted but not yet experienced, if a human operator is nearing the limits of her ability to stay awake, if a human operator experiences abnormal but not threatening health issues, etc.).

[0027] A vehicle may receive an indication from third-party sources of an impending road hazard and may determine the urgency of altering the vehicle's autonomous capabilities on its own. Vehicles may communicate wirelessly with one another in to relay updates to one another regarding changing road conditions as experienced by the vehicles themselves. The vehicles may communicate according to a peer-to-peer network in which the vehicles connect with one another directly (e.g., a swarm) or a centralized party may collect information from the vehicles, optionally process the information, and selectively provide relevant information to vehicles as the vehicles need it (e.g., client/server). For example, if a vehicle crash occurs on a road and vehicles in the vicinity detect the crash occurrence, the vehicle in the vicinity of the crash may communicate an emergency signal to vehicles approaching the crash site on the road such that those vehicles may take precautions to avoid crashing into any vehicles that are slow, stopped, or otherwise navigating differently than expected due to the crash.

[0028] In another implementation, a vehicle may receive an indication from a third-party that the vehicle should alter its autonomous capabilities and may receive a target time period in which to make the change therewith. A centralized authority (e.g., a weather prediction bureau, an insurer, a vehicle owner, a government agency, etc.) may determine that a change should be made to a vehicle's autonomous capabilities based on information regarding the vehicle's environment and may communicate a request to make the change to the vehicle. Such a request may accompany a target time in which to make the change to the vehicle's capabilities.

[0029] A vehicle may adjust its autonomous capabilities to increase or decrease the number of autonomous capabilities, depending on the situation the vehicle is in. In some situations, the vehicle maybe ill-suited to safely navigate a situation autonomously, and manual control may be preferable (e.g., navigation in close quarters with other vehicles wherein human communication is needed to coordinate with the operators of other vehicles, if the vehicle experiences a sensor or other hardware failure, if the vehicle loses a communications link, etc.). In other situations, a

human operator may be more ill-suited to safely navigate a situation than the vehicle would be autonomously. It may be known that human operators in general, or a specific human operator, is likely to make a mistake that could lead to a crash that an autonomously controlled vehicle would not make. For example, if a vehicle is traveling at a high rate of speed in low visibility conditions (e.g., heavy fog, frequent road elevation changes, blind spots, etc.) and is approaching a sudden traffic jam wherein other vehicles are traveling at a comparatively much slower rate of speed or are stopped on a road, it may be known that human operators are less likely to reduce speed in time than an autonomous vehicle would be. If the vehicle detects such a road condition approaching, it may request the human operator relinquish some or all control of the vehicle to reduce risk of a vehicle crash.

[0030] FIG. 1 is a schematic diagram of an example system **100** for detecting human operator parameters in a vehicle. The system **100** includes a vehicle **102** with more than one level of autonomous capabilities navigating on a road **104**. The vehicle **102** may include one level of pure manual control and one or more additional levels of enhanced automated control. When the vehicle approaches a potential road hazard **106** or other road conditions under which a change in the autonomous capabilities of the vehicle **102** are warranted, the vehicle **102** may display a notification to the human operator **110** to expect an upcoming or immediate change in the responsibilities of the human operator **110** in piloting the vehicle **102**.

[0031] One factor in determining the type of notification of a change in autonomous capabilities of the vehicle **102** to the human operator **110** is the current state of the human operator **110**. An alertness level of the human operator **110** may be estimated based on objective data collected by sensors inside the vehicle **102** such as the sensors shown in the bubble **108** of the interior of the vehicle **102**. In one implementation, an optical imaging device **112** (e.g., a camera) is located inside the vehicle **102** and is directed towards the human operator **110**. The camera **112** may capture images of the human operator **110** that may be analyzed to extract human operator parameters used to determine an alertness state of the human operator.

[0032] In at least one implementation, the human operator parameters include a body position of the human operator **110**. If the human operator **110** is sitting upright in the seat **114** and has her hands on or near a steering wheel, it is likely that the human operator **110** will be more responsive to a change in driving responsibilities than if the human operator **110** is reclined in the seat **114** without hands near the steering wheel. Images captured by the camera **112** may be analyzed by components of the vehicle **102** to determine whether the human operator is in a body position that indicates a greater or lower level of alertness. The camera **112** may capture a series of images of the human operator **110** (e.g., a video) that may be compared by components of the vehicle **102** to determine an activity level of the human operator **110**. For instance, if the human operator **110** is asleep, then she will likely exhibit a different movement patterns than a human operator **110** who is awake. Another type of objective data that may be collected by the camera **112** regarding the human operator's alertness and preparedness for accepting a change in driving responsibilities is the activity in which the driver is engaged. If analysis of images captured by the camera **112** indicate that the human operator **110** is holding a book or electronic device, for example, then the human operator **110** is more likely to experience a slower change of focus away from the book and to road conditions than a human operator **110** who is not holding a book or electronic device. The camera **112** may also capture images of the face of the human operator **110** to determine whether her eyes are open or closed, focused on the environment outside of the vehicle **102** or inside the vehicle **102** and other factors such as fatigue.

[0033] The human operator parameters may also include non-visual data collected from the interior of the vehicle **102**. In at least one implementation, non-visual data includes biometric data of the human operator (e.g., heart rate, breathing rate, body temperature, perspiration rate, etc.). Biometric data may be collected via the seat **114** in the vehicle **102** because the seat is in physical contact with the human operator **110**, which facilitates the collection of various types of biometric data. For

example, a sensor **116** may be embedded in the seat such that the sensor **116** can collect relevant data (e.g., a thermometer, a heart rate sensor, a breathing rate sensor, a perspiration sensor, etc.). [0034] Other types of sensors that not embedded in the seat **114** may also be used to collect non-visual data for the human operator parameters. In one implementation, an audio sensor **118** detects sounds inside the vehicle **102**. If a human operator is speaking to another occupant of the vehicle **102** or on a phone, the human operator's speech will be recognized by the audio sensor **118**. The human operator parameters may then reflect that the human operator **110** is awake and likely somewhat but not completely focused on the environment surrounding the vehicle **102**. The sensor **118** may also detect other types of sounds such as the human operator **110** moving inside the vehicle **102**, suggesting an activity level of the human operator **110**.

[0035] Human operator parameters may also include historical information regarding the human operator **110**. In one implementation, components of the vehicle **102** compile historical data regarding the human operator **110**. For example, the speed at which the human operator **110** responded to various notifications of an upcoming change to the vehicle's autonomous capabilities (e.g., notifications of various levels of intrusiveness, audio notifications, video notifications, haptic feedback notifications, etc.) As another example, the vehicle **102** may compile, how well the human operator **110** reacted to various types of road conditions and potential hazards (e.g., how often does the human operator **110** engage in emergency braking in various road conditions such as snow or rain, does the human operator **110** tend to lane drift at high speed, is the human operator **110** prone to panicked control movements in heavy traffic, etc.).

[0036] Historical information regarding the human operator **110** may also be received via a communications device at the vehicle **102** from a third-party. In one implementation, an insurer of the vehicle **110** compiles historical driving information regarding the human operator **110** such as vehicle crash history, vehicle usage history, and biographical data such as age and education level that may be used to determine an alertness level of the human operator **110**. For example, if the insurer knows that the human operator **110** usually operates the vehicle early in the morning and never at night, the insurer may determine that that human operator **110** is not accustomed to driving in the middle of the night. If the human operator **110** does drive one day in the middle of the night, then the insurer may determine that this poses a greater than usual safety risk for the particular human operator **110**. Other rules may also be applied based on historical data such as young drivers under age **18** pose a greater risk when driving late at night, in adverse weather conditions, when the audio sensor **118** detects multiple young people in the vehicle **102** at the same time, etc.

[0037] FIG. 2 is a time-series schematic diagram of an example system **200** for manual control re-engagement in an autonomous vehicle **206** with a human operator **202**. The vehicle **206** includes an image capture device **204** such as a camera. The camera **204** may be directed towards the face of the human operator **202** and images captured thereof. Components of the vehicle **206** may analyze the images of the face of the human operator **202** to identify characteristics of the human operator **202** that are relevant to the human operator parameters used to determine an alertness level of the human operator **202**.

[0038] The camera **204** may analyze various features of the human operator **202** to supply human operator parameters to the vehicle **206**. For example, in a time period before time **210**, the eyes of the human operator **202** are open and focused outside the window of the vehicle **206**. The camera **204** may capture a series of images of the face of the human operator **202** to detect other features such as rapidity of eye movement, dilation of eye pupils, blinking, etc.

[0039] In a time period after time **210**, the camera **204** may capture more images of the face of the human operator **202**. In a time period after time period **210**, the eyes of the human operator **202** are still open, but are focused on a book **214**. Images captured by the camera **204** may reflect that the eyes of the human operator **202** are no longer focused on the environment surrounding the vehicle **206**, but instead are focused on the book **214**. Images captured by the camera **204** may record features of the eyes of the human operator **202** such as eye movements indicating a speed at which

the human operator **202** is reading. Slower reading speeds and other eye movements may indicate a fatigue level of the human operator **202**.

[0040] If the eyes of the human operator **202** are no longer focused on the environment surrounding the vehicle, human operator parameters may include a level of distraction represented by a “score.” For example, if the human operator **202** is focused on a book **214**, it may be likely that the human operator is engaging in an extended period of perhaps intense concentration on the book **214**. The longer the human operator **202** focuses on the book **214**, the more likely she is to have a higher level of detachment from her environment. Such behavior may indicate that the human operator parameters should reflect a higher level of distraction. A security arbiter in the vehicle **206** may set a relatively longer period of time that would be expected before the human operator **202** responds to a notification of change to driving responsibility based on a higher distraction score in the human operator parameters. On the other hand, if the human operator **202** is only occasionally focusing on a handheld device (e.g., checking email, etc.), then the distraction of the human operator **202**, while still present, may not be considered as distracted as long periods of reading a book **214**. In such a case, a lower environmental detachment score may be included in the human input parameters and relied on by other components of the vehicle **206** to alert the human operator of an impending change in driving responsibility and management of shifting the vehicle from one level of autonomous control to another.

[0041] At a time period after time **212**, the camera **204** may capture additional images of the face of the human operator **202** that indicate the human operator **202** is suffering from fatigue. Images captured by the camera **204** may show that that eye lids of the human operator **202** are not as open as before the human operator **202** began experiencing fatigue. Other eye-based indications include blinking rate, location of eye focus, and eye movement rapidity.

[0042] The human operator parameters collected by the camera **204** are used to determine an alertness level of the human operator **202**. The determined alertness level may be used to determine whether a change should be made to the autonomous capability level of the **206** and whether the human operator **202** should take on more or less driving responsibility. If the vehicle **206** determines that an approaching road hazard **208** exists and the human operator **202** should take on more driving responsibility, then the vehicle may display a notification to the human operator **202**. The type of notification displayed to the human operator **202** and the time period for which the notification should be displayed in advance of an encounter with the road hazard **208** depend on the alertness level of the human operator **202** and the human operator parameters sensed by components of the vehicle **206**. For example, if the human operator **202** has a higher alertness level, such as in the time period before time **210**, then the notification of a change to the vehicle's autonomous capabilities may be less intrusive or occur closer to the road hazard **208** because it is expected that the human operator **202** will be able to recognize the notification and increase driving responsibility relatively quickly. On the other hand, if an alertness of the human operator **202** is lower due to human operator parameters such as those examples after time **210** and after time **212** (e.g., occupied with another task, experiencing fatigue, etc.), then the notification of a change to the vehicle's autonomous capabilities may be more intrusive or occur farther away from the road hazard **208** because it is expected that the human operator **202** will need relatively more time to recognize the notification and prepare to increase driving responsibility.

[0043] FIG. **3** is an in-vehicle view of an example system for manual control re-engagement **300** in an autonomous vehicle. Included in the system **300** are a number of types of notifications to alert a human operator that the vehicle may shift to a different level of autonomous capability. The notifications range from less intrusive to a human operator to more intrusive to a human operator, depending on the urgency of the shift to another autonomous vehicle capability level and/or any approaching road hazards. Notifications that the vehicle will shift to a different level of autonomous capability may be tailored to an individual human operator based on collected human operator parameters, such as whether the particular human operator is in an occupied or distracted

state, whether the particular human operator tends to respond well to certain types of notifications over other types of notifications, whether the human operator exhibits biomarkers that indicate a reduced capacity for driving (e.g., accelerated or irregular heart rate), whether an impending shift will require a greater or lesser degree of vehicle control on behalf of the human operator, etc.

[0044] In the implementation illustrated in FIG. 3, a text notification **302** appears on the dashboard of the vehicle. In one implementation, the text notification **302** alerts a human operator of an impending change to the level of autonomous control exercised by the vehicle. For example, if the vehicle or another entity has determined that a greater degree of autonomous control is better suited to the potential road hazard, then the text notification **302** may indicate to the human operator that the vehicle will include greater autonomous capability (and thus less human responsibility) to navigate the upcoming road hazard. The text notification may include information regarding the road hazard or an impending shift of autonomous vehicle capabilities (e.g., whether the human operator is expected to exercise a greater or lesser degree of control, the urgency of the hazard or shift in capabilities, an expected time until the hazard is encountered or the shift in capabilities is executed, etc.).

[0045] The text notification **302** may be adjusted to be more intrusive or less intrusive to the human operator, depending on the nature of the road hazard or change in autonomous capabilities. If the vehicle is preparing to gain more control over navigation of the vehicle with a corresponding reduction in responsibility on the part of the human operator, then the notification **302** need not be highly intrusive because the notification **302** is more of an informational nature. On the other hand, if there is a problem with the vehicle's autonomous capabilities and the human operator is expected to take some or all responsibility for piloting the vehicle, then the notification **302** may be more intrusive to the human operator because the human operator must be made aware of the alert, which could involve interrupting another activity that the human operator may be engaged in (e.g., sleeping, reading, working, in conversation with another occupant of the vehicle, etc.). The text notification **302** may be made more or less intrusive to the human operator by changing text font, flashing text, increasing or decreasing brightness of the text notification **302**, using colors to indicate urgency (e.g., red for high urgency, yellow or intermediate urgency, green for lower urgency, etc.)

[0046] In the implementation illustrated in FIG. 3, haptic feedback **304** may be used to notify a human operator of an impending road hazard or change in autonomous vehicle capabilities. Haptic feedback **304** may emanate from haptic devices embedded in various parts of the interior of the vehicle such as in the vehicle seats, arm rests, headrests, vehicle and/or media control surfaces, etc. Haptic feedback signals **304** may accompany a text notification **302**. Haptic feedback signals **304** may be used to make a notification more intrusive to a human operator. For example, if a human operator is deemed to be in a state of lowered attention span, such as when sleeping, reading a book, etc., then a haptic feedback signal **304** may assist in helping the human operator to “snap out” of her current state and to re-focus on the notification **302** and/or any road or vehicle conditions that require her attention.

[0047] Additionally, or alternatively, an audio alert **306** may be used to notify a human operator of an impending road hazard or impending change in autonomous vehicle capabilities. An audio alert **306** may emanate from speakers located around the interior of the vehicle. Audio alert **306** may include sounds designed to attract the attention of a human operator and/or voice commands directed to the human operator. For example, a voice command in the audio alert **306** may instruct the human operator to prepare for an increase in driving responsibility or signal to the human operator how much time is left until the human operator will be expected to exercise more or less control over the vehicle. The audio alert **306** may be made more intrusive or less intrusive to a human operator by varying the volume of the audio alert **306**, including a sound reflective of the urgency of impending changes to the human operator's responsibilities (e.g., an alarm sound for an urgent hazard, a soft voice or bell for a less urgent hazard, etc.).

[0048] In at least one implementation, other visual alerts **308** are included to notify a human operator of an impending road hazard or impending change in autonomous vehicle capabilities. In one implementation, the visual alerts **308** include a light bulb (e.g., a light emitting diode) for flashing or steady illumination. In other implementations, the visual alerts **308** include an e-fabric display of images or colors on the interior of the vehicle (e.g., an alert symbol, a text message, a color indicating urgency, etc.). Like other alerting mechanisms in the vehicle, the visual alert **308** may be adjusted to be more intrusive or less intrusive to the human operation by varying the intensity of the alert (e.g., brightness of a light bulb), flashing the visual alert **308**, changing the color of the visual alert **308**, changing a message or symbol displayed by the visual alert **308**, etc.

[0049] FIG. **4** is an in-vehicle view of an example system for manual control re-engagement **400** in a vehicle. The system **400** includes a text display **402** for alerting a human operator of an impending change to autonomous vehicle capabilities of the vehicle. In the implementation illustrated in FIG. **4**, the vehicle or another party has determined that an expected road hazard should not be navigated by the vehicle autonomously. Some road hazards (e.g., a rock slide) are very serious and seldom encountered by autonomous vehicle. It may not be known how well an autonomous vehicle is able to navigate such a hazard due to unpredictable conditions and infrequent collection of the behavior of autonomous vehicle in such a situation. In such cases, it may be determined that a human operator must quickly assume control of the vehicle.

[0050] The determination that an expected road hazard should not be navigated autonomously may be based on information collected by the vehicle itself or may be reported to the vehicle via a communications interface (e.g., other vehicles in the area may send reports that they are not successfully navigating a road hazard, an insurer may determine that human operators in general or a human operator in particular is more likely to safely navigate a road hazard than a vehicle under autonomous control, etc.). The text notification **402** may be accompanied by other notification methods (e.g., audio signal, haptic feedback, visual signals, etc.). The text notification may further include information such as an expected amount of time until a road hazard will be reached or an expected amount of time until a change to autonomous driving capabilities takes effect.

[0051] FIG. **5** is an in-vehicle view of an example system for manual control re-engagement **500** in a vehicle. The system **500** includes a text notification **502**. Like the text notifications disclosed herein with reference to other implementations, the text notification **502** includes a message to the human operator regarding an impending road hazard and/or an impending change to the autonomous capabilities of the vehicle. Text notification **502** may include a disruption to other content displayed inside the vehicle such as presentation of video, entertainment information (e.g., video games, web surfing, books, etc.), maps, etc. Interruption of other content may make the notification **502** more intrusive into the human operator's environment, which may be appropriate depending on the severity of the alert to be displayed.

[0052] The system **500** may include a warning alert **504** that may be displayed at various points around the interior of the vehicle. In at least one implementation, a warning alert **504** is displayed in locations wherein it is likely to attract the attention of a human operator. For example, the warning alert **504** may be displayed on a rear-view mirror, on the steering wheel, in a heads-up display on the windshield, etc. Whether a warning alert **504** is to be displayed depends in part on an urgency of attracting the human operator's attention based on the human operator parameters and alertness level of the human operator.

[0053] FIG. **6** is an in-vehicle view of an example system for manual control re-engagement **600** in a vehicle. In the implementation illustrated in FIG. **6**, a notification **602** includes a request to the human operator to adjust the level of autonomous control exercised by the vehicle. In some situations, the vehicle may determine or a third party may determine, that a vehicle is likely to be navigated more safely autonomously than by the human operator. Examples of third party entities that may make such a determination include a vehicle manufacturer, a government agency, an insurer, a vehicle owner, etc. The vehicle or the third parties may also determine that a vehicle is

likely to be operated more safely manually than autonomously in certain situations. In the example illustrated in FIG. 6, an extreme weather alert issued from a weather service initiates a request to the human operator to agree to allow the vehicle reduce or eliminate autonomous capabilities in favor of manual control. A human operator may interact with the notification **602** directly or by other controls in the interior of the vehicle to accept the request presented in notification **602**.

[0054] FIG. 7 is an in-vehicle view of an example system for manual control re-engagement **700** in an autonomous vehicle. In at least one implementation, an insurer determines whether a human operator or a vehicle is more likely to safely navigate the vehicle in certain conditions. The insurer initiates a request to the human operator to be displayed as notification **702**. The notification **702** may include an offer from an insurer to lower insurance premium prices in return for the human operator's agreement to allow the vehicle to assume a greater or lesser degree of autonomous control over road navigation depending on the rules preferred by the insurer. Rules preferred by the insurer may depend on factors such as the human operator parameters measured inside the vehicle, a driving history of the particular human operator in the vehicle, and data gathered regarding events outside the vehicle.

[0055] FIG. 8 is a schematic diagram **800** of a vehicle **802** equipped with an example system for manual control re-engagement in an autonomous vehicle approaching a road hazard **804** on a road **806**. Under some circumstances, it may be safer for the vehicle **802** to alter available autonomous capabilities to more safely navigate the road hazard **804**. To make a determination whether it is safer to alter autonomous vehicle capabilities or to remain in the status quo, the vehicle detects a variety of human operator parameters regarding a human operator of the vehicle to determine a human operator alertness level.

[0056] Also relevant to a determination of whether it is safer to alter autonomous vehicle capabilities of the vehicle **802** is an assessment of conditions on the road **806** and of the road hazards **804**. In at least one implementation, the vehicle **802** receives information regarding conditions on the road **806** and of the road hazard **804** from other vehicles and from sensors in the road **806** itself.

[0057] The vehicle **802** includes components for managing a transition from one level of autonomous capabilities to another level of autonomous capabilities. One type of component to manage the transition is an evaluator in the vehicle **802** to evaluator to evaluate readiness of a human operator of a vehicle including one or more sensors. A security arbiter in the vehicle **802** determines a security risk to the vehicle (e.g., a road hazard, adverse conditions, diminished capacity of a human operator, etc.) and determines whether a level of autonomous capability of the vehicle should be adjusted by adding or removing autonomous capabilities and conversely adding or removing driving responsibility from a human operator. The security arbiter in the vehicle **802** may receive human input parameters directly from the human operator and/or from a remote party.

[0058] In at least one implementation, the security arbiter in the vehicle **802** evaluates a threat posed by the road hazard **804**. The security arbiter may receive information regarding the threat from other road participants and remote parties. For example, vehicles **808** traveling in the same direction as the vehicle **802** but closer to the road hazard **804** may wirelessly relay information back to vehicle **802** and the other vehicles approaching the road hazard **804** on the road **806** behind the vehicles **808**. The security arbiter may receive indications, for example without limitation that the vehicles **808** have encountered sharply lower road speeds or have observed adverse road conditions. The vehicles **808** may further relay information to the vehicle **802** regarding the location of the road hazards **804** on the road **806**. The vehicles **808** may include telematics data in the information sent to the security arbiter in the vehicle **802** (e.g., heavy braking, high G-forces, etc.) As another example, third parties (e.g., government agencies, insurers, vehicle owners, etc.) may collect information regarding the conditions on the road **806** by way of remote sensors **812**. The remote sensors **812** may detect road speeds and be able to determine whether vehicles are in distress or if a dangerous condition has developed on the road **806**. Remote sensors **812** may be

fixed in place roadside (e.g., vibration sensor, vehicle counter, camera, etc.) or may be mobile sensors (e.g., drone, unmanned aerial vehicle, helicopter, etc.).

[0059] After receiving data regarding the road hazard **804**, the security arbiter may determine a security threat to the vehicle based on the road hazard **804** and other information available to the security arbiter. The security arbiter may further determine a change to autonomous vehicle capabilities that should be taken to improve safety when navigating the road hazard **804**. Such a determination may be based on information available to the security arbiter or it may be an instruction received from a third party. The security arbiter may further determine a time period during which the change in the vehicle's autonomous capability should occur.

[0060] Another component in the vehicle **802** is a notification server. The notification server in the vehicle **802** may include hardware and software components (e.g., a display for text messages to a human operator inside the vehicle, speakers for playing audio text notifications and instructions, lights, feedback devices, an operating system, microphones, etc.) for presenting information to and receiving information from a human operator of the vehicle **802**. The notification server presents notifications to a human operator of the vehicle **802**. In at least one implementation, the security arbiter determines a time period and an intrusiveness level of

[0061] After receiving data regarding the road hazard **804**, the security arbiter may select a time period during which the vehicle should transition to a different level of autonomous capability. The time period may be calibrated based on the speed of the vehicle **802** on the road **806**, the location of the road hazards **804**, information received from vehicles **808**, etc. In at least one implementation, the security arbiter may decrease speed of the vehicle to lengthen the time until the vehicle **802** reaches the road hazard **804**. In at least one implementation, the security arbiter provides a time period to the notification server of the vehicle **802** during which changes to the vehicle's autonomous capabilities should be made. In implementations, a security arbiter requests the notification server to require human operator acknowledgement before implementing changes to the autonomous capabilities of the vehicle **802**. In another implementation, the security arbiter will cease navigation of the vehicle if the human operator does not acknowledge increased driving responsibilities. In other implementations, the security arbiter implements a change in the autonomous capabilities of the vehicle **802**.

[0062] FIG. **9** is an in-vehicle view of an example system **900** for manual control re-engagement in a vehicle. The system **900** includes a text notification **902** including request for acknowledgement from the human operator. The text notification **902** may serve as a “ping” to the human operator to test for responsiveness. When a human operator responds to the text notification **902** (e.g., through a touch interface, pushing a button, speaking a response into a microphone, etc.), the system **900** (e.g., a security arbiter on the vehicle) may record a responsiveness time associated with the human operator's acknowledgement of the text notification **902**. An intrusiveness level of the text notification **902** may be varied to test the human operator's reaction to different levels of intrusiveness. Some human operators may react well to minimally intrusive notifications and find more intrusive notifications to be irritating. Other human operators, on the other hand, may not respond well to minimally intrusive notifications and may need more intrusive notifications to respond to a request by the vehicle to assume more driving responsibility.

[0063] FIG. **10** illustrates example operations **1000** for requesting engagement of manual control in a vehicle. An identifying operation **1002** identifies a human operator of a vehicle, the human operator being associated with a driving capability profile. The identifying operation **1002** may rely on sensors inside the vehicle to identify the human operator (e.g., facial recognition) or the identifying operation may identify the human operator of the vehicle via credentials (e.g., bio credentials, answering a cryptographic challenge, username and password, possession of a physical vehicle key, etc.). The driving capability profile may be formed from data collected regarding the human operator. The data regarding the human operator may include driving history data such as previous reactions to road hazards, previous vehicle crashes, data regarding the type of

notifications the human operator tends to respond well to, the amount of time the human operator typically needs to respond to a notification, the amount of time the human operator typically needs to acknowledge an acknowledgement request, types of road conditions under which the human operator typically exhibits good or poor driving skills, the frequency with which the human operator becomes intoxicated or otherwise experiences diminished driving capabilities, the human operator's schedule, whether the human operator is likely to be sleep-deprived, etc.

[0064] A determining operation **1004** determines human operator parameters associated with the human operator. The determining operation **1004** may include sensors inside the vehicle to collect biometric data regarding the human operator (e.g., camera, heart rate sensor, perspiration sensor, steering wheel grip force sensor, body temperature sensor, movement sensors, microphones, etc.). The determining operation **1004** may also rely on information about the human operator stored by the vehicle or communicated to the vehicle such as demographic information regarding the human operator (age, gender, length of driving experience, etc.) and historical information regarding the human operator's usage of the vehicle.

[0065] Another determining operation **1006** determines an alertness level of the human operator based on the human operator parameters and the driving capability profile. The determining operation **1006** may include a determination of an amount of time the human operator is likely to need to respond to notifications of various levels of intrusiveness.

[0066] A displaying operation **1008** displays a manual control request notification to the human operator. The manual control request notification may be based on the determined alertness level of the human operator. If the determining operation **1006** determines that a human operator has a low alertness level, then the displaying operation **1008** may display a more intrusive notification to the human operator and may display the notification earlier than if the human operator has a high alertness level. If the determining operation **1007** determines that a human operator has a high alertness level and the driving capability profile and/or the human operator parameters indicate that the human operator responds well to a certain type of notification, then the displaying operation **1008** may display that notification to the human operator.

[0067] A transferring operation **1010** transfers control of the vehicle to the human operator if the human operator accepts the request for manual control displayed in operation **1008**. In one implementation, a human operator accepts the request for manual control by interfacing with an input such as a touch interface or control inside the vehicle. In another implementation, the human operator accepts the request for manual control by moving the vehicle controls (e.g., steering wheel, brake pedal, etc.).

[0068] FIG. **11** illustrates example operations **1100** for transitioning control of a vehicle. A navigating step **1102** navigates a vehicle according to a first level of autonomous capability. The first level of autonomous capability includes a set of autonomous competencies exercised by the vehicle while it is operating according to the first level of autonomous capability. Autonomous capabilities include without limitation adaptive cruise control, parking assist (steering only), lane assist, full parking assist, traffic jam assist, conditional automation, high automation, and full automation. These autonomous capabilities may be organized according to ascending levels (e.g., Levels 0-5) as the autonomous competencies take over more and more driving responsibilities from a human operator. A vehicle navigating according to a first level of autonomous capability exercise some (or all) of these autonomous competencies, depending on which autonomous competencies are included in the first level. For example, if the first level of autonomous capability is Level 3 autonomous capability, then a vehicle navigating according to the first level of autonomous capability may exercise lane assist and parking assist, but not conditional automation.

[0069] A receiving operation **1104** receives a request to operate the vehicle according to a second level of autonomous capability. The receiving operation **1104** may receive the request from an entity separate from the vehicle itself. For example without limitation, a government agency monitoring road conditions, a vehicle manufacturer, an insurer, a vehicle owner, etc. may send a

request to the vehicle to operator according to a second level of autonomous capability. In another implementation, the vehicle itself may determine that navigation is likely to be safer if the vehicle shifts to a different level of autonomous capability. For example, a security arbiter on the vehicle may evaluate human operator parameters and a human operator alertness level in comparison to any known or expected road hazard to create the request to operate the vehicle according to the second level of autonomous capability, and the vehicle may perform the receiving operation **1104** in response to the security arbiter's request.

[0070] A notifying operation **1106** notifies a human operator with a notification of an impending change to the vehicle's autonomous capabilities. In one implementation, the notifying operation **1106** is performed by a notification server on the vehicle. The notification may be adjusted by the notification server to be more intrusive or less intrusive to the human operator depending on the information conveyed therein. For example, if the impending change in vehicle capability includes a reduction to the driving responsibilities of the human operator, a less intrusive notification may be selected. On the other hand, if the impending change in vehicle capability includes an increase in driving responsibility of the human operator, then a more intrusive notification may be used. A navigating step **1108** navigates the vehicle according to the second level of autonomous capability. [0071] Of course, the applications and benefits of the systems, methods and techniques described herein are not limited to only the above examples. Many other applications and benefits are possible by using the systems, methods and techniques described herein.

[0072] Furthermore, when implemented, any of the methods and techniques described herein or portions thereof may be performed by executing software stored in one or more non-transitory, tangible, computer readable storage media or memories such as magnetic disks, laser disks, optical discs, semiconductor memories, biological memories, other memory devices, or other storage media, in a RAM or ROM of a computer or processor, etc.

[0073] The patent claims at the end of this patent application are not intended to be construed under 35 U.S.C. § 112(f) unless traditional means-plus-function language is expressly recited, such as “means for” or “step for” language being explicitly recited in the claim(s). The systems and methods described herein are directed to an improvement to computer functionality, and improve the functioning of conventional computers.

Claims

1. A method of transitioning control of a vehicle, the method comprising: navigating a vehicle according to a first level of autonomous capability, the first level of autonomous capability defining a first set of autonomous competencies of the vehicle; notifying a human operator of the vehicle with a notification of an impending change in vehicle capability to a second level of autonomous capability, the notification having a user intrusiveness level based at least in part on a change in proximity of a hazard; and navigating the vehicle according to the second level of autonomous capability.
2. The method of claim 1, further comprising: receiving an indication that the hazard is impending; and determining the impending change in vehicle capability is urgent responsive to receiving the indication.
3. The method of claim 2, further comprising: receiving the indication from a third-party source.
4. The method of claim 3, wherein the third-party source is another vehicle.
5. The method of claim 3, wherein the third-party source is a sensor within or proximate a roadway being traveled by the vehicle.
6. The method of claim 2, further comprising: determining a time period for which the notification is displayed in advance of an encounter with the hazard based on an alertness level of the human operator.
7. The method of claim 2, further comprising: determining a time period for which the notification

is displayed in advance of an encounter with the hazard based on a driving history of the human operator.

8. The method of claim 2, wherein: the notification includes a text notification, and the text notification indicates an expected time until the hazard is encountered.

9. The method of claim 2, wherein: the notification indicates that greater autonomous capability is desired to handle the hazard.

10. The method of claim 2, wherein: the notification includes an audible notification, and the audible notification indicates an expected time until the hazard is encountered.

11. A system for transitioning control for a vehicle, the system comprising: a vehicle; a control system for implementing autonomous control of the vehicle, the control system configured to: navigate a vehicle according to a first level of autonomous capability, the first level of autonomous capability defining a first set of autonomous competencies of the vehicle; notify a human operator of the vehicle with a notification of an impending change in vehicle capability to a second level of autonomous capability, the notification having a user intrusiveness level based at least in part on a change in proximity of a hazard; and navigate the vehicle according to the second level of autonomous capability.

12. The system of claim 11, wherein the control system is further configured to: receive an indication that the hazard is impending; and determine the impending change in vehicle capability is urgent responsive to receiving the indication.

13. The system of claim 12, wherein the control system is further configured to: receive the indication from a third-party source.

14. The system of claim 13, wherein the third-party source is another vehicle.

15. The system of claim 13, wherein the third-party source is a sensor within or proximate a roadway being traveled by the vehicle, and wherein the system further includes wireless communication circuitry to receive sensor information from within or proximate the roadway.

16. The system of claim 12, wherein the control system is further configured to: determine a time period for which the notification is displayed in advance of an encounter with the hazard based on an alertness level of the human operator.

17. The system of claim 12, wherein the control system is further configured to: determine a time period for which the notification is displayed in advance of an encounter with the hazard based on a driving history of the human operator.

18. The system of claim 12, wherein: the notification includes a text notification, and the text notification indicates an expected time until the hazard is encountered.

19. The system of claim 12, wherein: the notification indicates that greater autonomous capability is desired to handle the hazard.

20. The system of claim 12, wherein: the notification includes an audible notification, and the audible notification indicates an expected time until the hazard is encountered.
